[Subject Code-2610000102066001]

Emerging Technologies and Applications

FYBBA SEMESTER -II SEC201	Emerging Technologies and Applications	1L:0T:2P	2 Credits

Course Objective:

- To provide a comprehensive understanding of emerging technologies such as block chain, IoT, cloud computing, robotics, AR/VR, etc.
- To explore the applications, implications, and strategic advantages of emerging technologies in business for competitive advantage.

Contents:

Unit 1: Cloud Computing

Cloud Service Models: Infrastructure as a Service (IaaS), Platform as a Service (PaaS), Software as a Service (SaaS), Deployment Models: Public, Private, Hybrid, Cloud-Based Solutions: use of cloud in business (Enterprise solutions), . Benefits & Challenges of Cloud Computing

Unit 2: Internet of Things (IoT) & Industry 4.0

IoT Applications: Smart cities, infrastructure, industrial IoT, manufacturing, Data processing and storage. Industry 4.0: Concept, automation, smart manufacturing, cyber-physical systems, digital twins.

Unit 3: Blockchain Technology

What is blockchain technology, Basic components of blockchain technology, How decentralization and shared ledgers work, Applications of Blockchain Technology, advantages and disadvantages of Blockchain Technology.

Unit 4: Augmented Reality (AR) and Virtual Reality (VR)

- Introduction to AR/VR and differences, AR/VR applications in marketing and Enhancing customer experience., Technological limitations and advancements.

Practical (Suggestive List):

- Hands on sessions on utilizing popular cloud platforms for development and deployment, offering hands-on experience with free tiers and trial accounts.
- Hands on sessions on block chain technologies, focusing on the basics development and deployment of decentralized applications.

Readings:

Text Books (Latest Editions):

1. Emerging Technologies by Errol S. van Engelen
2. Internet of Things by Jeeva Jose, Khanna Book Publishing.
3. Digital Transformation: A Strategic Approach to Leveraging Emerging Technologies, Anup Maheshwari
4. Virtual & Augmented Reality by Rajiv Chopra, Khanna Book Publishing.

Case Studies

1. Software and/or Data: Dilemmas in an AI Research Lab of an Indian IT Organization, Rajalaxmi Kamath; Vinay V Reddy, https://hbsp.harvard.edu/product/IMB889-PDF-ENG?Ntt=emerging%20technologies
2. Volkswagen Group: Driving Big Business With Big Data, Ning Su; Naqaash Pirani, https://hbsp.harvard.edu/product/W14007-PDF-ENG?Ntt=emerging%20technologies

Course Outcomes:

1. Students will **understand** foundational knowledge of emerging technologies such as blockchain, IoT, cloud computing, AR/VR, etc., comprehending their principles, components, and functionalities.
2. Students will **analyze** the practical applications of these technologies in various business contexts, evaluating how they can optimize operations, enhance decision-making, and drive innovation.
3. Students will **evaluate** the strategic implications of adopting emerging technologies, including potential challenges, risks, and opportunities, to formulate informed strategies for competitive advantage.
4. Students will develop skills to plan and manage the integration of emerging technologies into business processes, ensuring alignment with organizational goals and effective change management.

----

5. Emerging Technologies for Effective Management by Rahul Dubey, Cengage Publications.
6. IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things by David Hanes, Jerome Henry, Rob Barton, Gonzalo Salgueiro and Patrick Grossetete.
7. Blockchain for Business by Jai Singh Arun, Jerry Cuomo and Nitin Gaur.
8. Block Chain & Crypto Currencies by Anshul Kausik, Khanna Book Publishing.
9. Industry 4.0 Technologies for Business Excellence: Frameworks, Practices, and Applications by Edited By Shivani Bali, Sugandha Aggarwal, Sunil Sharma.
10. Blockchain, Artificial Intelligence, and the Internet of Things: Possibilities and Opportunities" by Pethuru Raj, Ashutosh Kumar Dubey, Abhishek Kumar, Pramod Singh Rathore.

Readings:

- Abdi, S., Kitsara, I., Hawley, M. S., & de Witte, L. P. (2021). Emerging technologies and their potential for generating new assistive technologies. Assistive Technology, 33(sup1), 17–26. https://doi.org/10.1080/10400435.2021.1945704
- Seokbeom Kwon, Xiaoyu Liu, Alan L. Porter, Jan Youtie, Research addressing emerging technological ideas has greater scientific impact, Research Policy, Volume 48, Issue 9, 2019, 103834, https://doi.org/10.1016/j.respol.2019.103834.
- Philip, J. (2022), "A perspective on embracing emerging technologies research for organizational behavior", Organization Management Journal , Vol. 19 No. 3, pp. 88-98. https://doi.org/10.1108/OMJ-10-2020-1063